

Simulation Scenario Standard - (SSS)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Simulation Scenario Standard (SSS) defines the governance conditions under which simulation scenarios are identified, selected, structured, varied, bounded, stressed, combined, prioritized, documented, and assessed for sufficient coverage so that a governed simulation examines not only expected operating conditions but also materially relevant variations, edge cases, stress conditions, uncertainty ranges, alternative states, transitions, failures, interactions, and plausible combinations capable of affecting simulation outcomes.

SSS establishes the scenario-governance layer through which the governed Simulation Object, Purpose & Scope, Reality Representation, Assumptions, Model & Environment, and Input & Data Basis are transformed into an explicit set of conditions that the simulation will actually examine.

It ensures that a technically capable simulation cannot create an appearance of adequate examination merely because it executes many runs while materially relevant scenarios remain absent.

Core Governance Question

Which scenarios must be simulated to represent expected conditions, variations, edge cases, stresses, uncertainty, and materially relevant alternatives?

Purpose

A simulation can possess: a correctly identified Simulation Object, a clear purpose and scope, a strong representation of reality, governed assumptions, a well-constructed model and environment, and suitable input data,

and still fail to provide useful simulation insight because the wrong scenarios were selected.

A simulation may test only:

normal conditions

while the real risk exists in:

degraded conditions.

It may test:

individual failures

while the critical problem arises from:

combined failures.

It may test:

average behavior

while material consequences occur at:

distribution tails.

It may test:

steady-state operation

while the dangerous condition appears only during:

transition.

SSS exists to govern this problem.

Operational Role

SSS is the seventh Parent Standard of SIMULOS® — Simulation Governance Architecture.

The preceding Parent Standards establish:

- 1. Simulation Object — What exactly is being simulated?**
- 2. Simulation Purpose & Scope — Why is the simulation being performed and what must it cover?**
- 3. Reality Representation — Which aspects of reality must be represented and at what abstraction and fidelity?**
- 4. Simulation Assumptions — Which premises enable the simulation and under what limitations?**
- 5. Simulation Model & Environment — Through which controlled model and environment will the simulation operate?**
- 6. Simulation Inputs & Data — Which information may enter and influence the simulation?**

SSS asks:

Which governed combinations of conditions must that simulation actually examine?

It therefore establishes the scenario basis required before governed Simulation Execution begins.

Governance Flow

Governed Simulation Object



Simulation Purpose & Scope



Reality Representation



Governed Assumptions



Governed Model & Environment



Governed Input & Data Basis



Scenario Requirement Identification



Baseline & Expected Conditions



Scenario Variation



Edge-Case Identification



Stress-Condition Design



Scenario Combination & Interaction



Scenario Coverage Assessment



Scenario Acceptance / Expansion / Restriction



Governed Simulation Scenario Set



Simulation Execution

Operational Space

SSS governs simulation-specific conditions relating to: scenario identification, scenario purpose, scenario identity, scenario classification, baseline scenarios, nominal scenarios, expected-condition scenarios, alternative scenarios, scenario variation, parameter variation, input variation, environmental variation, behavioral variation, uncertainty-driven scenarios, boundary scenarios, edge cases, rare conditions, failure scenarios, degraded-state scenarios, transition scenarios, recovery scenarios, stress conditions, extreme conditions, combined conditions, correlated conditions, multi-factor scenarios, cascading scenarios, sequential scenarios, scenario interaction, scenario dependency, scenario prioritization, scenario exclusion, scenario sufficiency, scenario coverage, scenario gaps, scenario redundancy, scenario comparability, scenario change, scenario versioning, and scenario-to-execution traceability.

Operational Scope

SSS applies wherever a simulation must intentionally determine which conditions are to be examined.

It may govern scenario design for: AI systems, autonomous systems, autonomous agents, robotics, aerospace systems, spacecraft, mission systems, industrial systems, manufacturing, critical infrastructure, transportation, digital twins, financial systems, energy systems, healthcare systems, scientific models, climate and environmental systems, cyber systems, defense systems, emergency systems, organizational processes, human-machine systems, multi-agent environments, and future operational ecosystems.

SSS does not prescribe which domain-specific scenarios are universally correct.

It governs how a simulation establishes that the scenarios selected for its specific purpose are sufficiently structured and sufficiently complete.

Simulation Scenario

A Simulation Scenario is a governed configuration of materially relevant conditions, states, inputs, parameters, events, behaviors, interactions, constraints, or transitions under which a Simulation Object is examined through simulation.

Conceptually:

Scenario = Defined Simulation Conditions + Applicable Configuration + Relevant Variation + Execution Intent

A scenario may describe: one operating condition, one failure, one environmental state, one behavior pattern, one combination of conditions, one transition, one stress state, or a complete event sequence.

Scenario Identity

Every material scenario should be sufficiently identifiable.

A Scenario Identity may establish: Scenario Identifier, scenario name, scenario purpose, relevant Simulation Object, applicable Simulation Scope, model and environment configuration, relevant Input Set, parameter configuration, environmental conditions, assumptions, scenario state, intended variation, and scenario version.

Scenario Requirement

Before scenarios are selected, SSS establishes Scenario Requirements.

A Scenario Requirement answers:

What condition, variation, boundary, uncertainty, interaction, or potential failure must the simulation be capable of examining?

The requirement exists independently of the exact technical scenario used to satisfy it.

This distinction supports traceability:

Scenario Requirement → Designed Scenario → Simulation Execution Baseline Scenario

A Baseline Scenario represents the governed reference condition against which materially different scenarios may be compared.

Baseline does not mean:

correct reality

or:

most likely future.

It establishes a controlled reference.

Nominal Scenario

A Nominal Scenario represents expected or intended operating conditions.

Nominal conditions may be important for: normal behavior, expected performance, baseline comparison, and operational-state understanding.

But nominal scenarios alone rarely establish sufficient scenario coverage for complex or high-consequence systems.

No Nominal-Only Principle

SIMULOS® establishes:

Successful simulation under nominal conditions does not demonstrate adequate simulation coverage when materially relevant non-nominal conditions exist.

Expected-Condition Scenario

Expected-condition scenarios represent conditions reasonably anticipated within the intended Simulation Scope.

They may include variation within: normal operating ranges, known environments, expected populations, expected loads, routine disturbances, or normal operational transitions.

Scenario Variation

Scenario Variation governs deliberate changes to one or more materially relevant simulation conditions.

Variation may involve: parameters, inputs, environmental conditions, system states, actor behavior, timing, resource availability, communication conditions, failures, or dependencies.

Controlled Scenario Variation

Where comparative interpretation matters, the simulation should identify which conditions are intentionally changed and which remain controlled.

Conceptually:

Baseline Scenario

↓

Change X

↓

Variation Scenario

This supports determination of whether differences in behavior are associated with the intended variation rather than hidden scenario changes.

Single-Factor Variation

A scenario may vary one material factor while maintaining other conditions.

This may help isolate sensitivity or causal contribution within the

simulation.

Multi-Factor Variation

Some behavior emerges only when several factors change simultaneously.

SSS therefore allows scenarios involving:

Factor A + Factor B + Factor C

where combined conditions are materially relevant.

No Single-Failure Sufficiency Principle

SIMULOS® establishes:

Testing individual failures does not establish coverage of combinations of failures where combined failures can produce materially different system behavior.

Edge Case

An Edge Case is a scenario representing conditions near or beyond the normal operating distribution, common expectation, typical configuration, or frequently observed state where simulation behavior may materially differ from nominal behavior.

An edge case may arise from: unusual inputs, rare environmental conditions, uncommon interactions, boundary values, atypical behavior, unusual sequencing, unexpected combinations, or low-frequency events.

Edge ≠ Impossible

A condition should not be excluded merely because it is rare.

The relevant governance question is:

Is the condition materially plausible and consequential within the Simulation Purpose & Scope?

Rare Scenario

A Rare Scenario may have low occurrence probability while retaining high consequence.

SSS therefore distinguishes:

Frequency

from:

Scenario Importance.

Consequence-Weighted Scenario Selection

Scenario selection should consider not only:

How likely is this condition?

but also:

What happens if it occurs?

A low-frequency catastrophic condition may warrant stronger scenario coverage than a high-frequency trivial variation.

Boundary Scenario

A Boundary Scenario examines behavior close to an applicable operating, environmental, physical, behavioral, or governance boundary.

Examples may involve: maximum load, minimum power, communication limits, environmental tolerance, control limits, resource exhaustion, operational envelopes, or response thresholds.

Boundary Crossing Scenario

Where materially relevant, scenarios should examine not only operation near a boundary but also what occurs when the boundary is crossed.

Conceptually:

Inside Boundary → Boundary Reached → Boundary Exceeded

The system may behave differently at each state.

Stress Scenario

A Stress Scenario intentionally subjects the simulated system to conditions beyond ordinary operating variation to determine how behavior changes under increased pressure or constraint.

Stress may involve: load, resource scarcity, environmental extremes, communication degradation, timing pressure, component degradation, information loss, actor density, adversarial behavior, or other system-specific stressors.

Stress Does Not Mean Arbitrary Extremity

Stress scenarios should remain relevant to the Simulation Purpose.

SSS does not require testing arbitrary impossible conditions merely to maximize scenario count.

Extreme Scenario

An Extreme Scenario represents a materially severe condition at or beyond ordinary expectations where system behavior or consequences may change significantly.

Extreme scenarios may be appropriate where: consequence is high, operational resilience matters, safety margins matter, rare-event behavior matters, or failure containment is being examined.

Failure Scenario

A Failure Scenario represents the failure of a component, dependency, capability, resource, interface, actor, process, or environmental condition relevant to the Simulation Object.

Failures may be:

Single

Multiple

Sequential

Simultaneous

Dependent

Cascading

or:

Common-Cause

where those distinctions matter.

Common-Cause Scenario

Multiple failures may arise from one common underlying cause.

For example:

Power Loss

may simultaneously affect:

Communications + Sensors + Cooling + Control

Treating the failures as statistically independent may miss the actual scenario structure.

Cascading Scenario

A Cascading Scenario examines a sequence in which one material event causes or enables another.

Conceptually:

Failure A → Degradation B → Resource Loss C → Failure D

Complex systems may fail through chains rather than isolated events.

Sequential Scenario

The order of conditions may materially change the outcome.

Therefore:

$A \rightarrow B \rightarrow C$

may require separate examination from:

$C \rightarrow B \rightarrow A$.

Transition Scenario

A system may behave safely in State A and safely in State B while becoming vulnerable during the transition between them.

A Transition Scenario therefore examines changes such as: startup, shutdown, mode change, handover, control transfer, autonomy escalation, environment transition, recovery, docking, landing, or state switching.

No Stable-State-Only Principle

SIMULOS® establishes:

Testing stable states alone does not establish coverage of transitions between those states where transition behavior may materially affect outcomes.

Degraded-State Scenario

A Degraded-State Scenario examines operation where the system remains functional but one or more capabilities, resources, dependencies, or operating conditions are reduced.

Examples may involve: reduced sensing, reduced communication, degraded power, limited computation, partial actuator availability, incomplete information, reduced human support, or degraded infrastructure.

Recovery Scenario

A Recovery Scenario examines behavior after disruption or failure.

It may address: restoration, restart, failover, fallback, state reconstruction, resource recovery, human intervention, or return to nominal operation.

SSS governs selection of the scenario.

ORA™ may govern actual Operational Recovery in real-world operation.

Uncertainty-Driven Scenario

Uncertainty identified in upstream SIMULOS® spaces may require deliberate scenario representation.

For example:

Input Uncertainty Range

may generate:

Low → Central → High

scenario conditions.

Uncertainty should not remain documented while scenario design tests only one convenient point estimate.

Unresolved-Uncertainty Scenario

Where uncertainty cannot be sufficiently quantified, scenario design may require alternative plausible states rather than one probabilistic representation.

Scenario Interaction

Scenarios may interact with: Simulation Assumptions, Input Sets, Environment Configurations, Parameters, actor behavior, dependencies, and execution conditions.

These relationships should remain explicit.

Scenario Dependency

A scenario may depend upon another condition being active.

For example:

Emergency Power Failure Scenario

may become relevant only after:

Primary Power Failure.

Scenario dependencies should remain traceable where material.

Scenario Combination

Individual scenario elements may need to be combined.

Conceptually:

Extreme Weather + Sensor Degradation + Communication Delay

may produce behavior not observable when each condition is simulated separately.

Combination Explosion

Complex systems can generate an enormous number of possible scenario combinations.

SSS does not require brute-force simulation of every mathematically possible combination.

It requires governed determination of which combinations are materially relevant.

Scenario Prioritization

Where exhaustive simulation is impractical, scenarios may be prioritized according to factors such as: materiality, consequence, probability, uncertainty, boundary proximity, dependency concentration, scenario novelty, known failure history, operational importance, and unresolved governance gaps.

Prioritization must not silently erase excluded material scenarios.

Scenario Selection Basis

Each material scenario or scenario family should have an identifiable selection basis.

This may include: Simulation Purpose, Reality Representation, Assumptions, uncertainty, input limitations, model sensitivities, operational states, known failure modes, boundary conditions, historical events, or defined exploratory objectives.

Scenario Exclusion

A potential scenario may be excluded where it is: outside Simulation Scope, materially impossible, non-relevant, duplicative, already sufficiently covered, or otherwise methodologically unnecessary.

Material exclusions should remain visible where omission could later be misunderstood as coverage.

No Convenience Exclusion Principle

SIMULOS® establishes:

A materially relevant scenario must not be excluded solely because it is difficult, computationally expensive, inconvenient, or likely to produce an undesirable result.

Scenario Feasibility

Scenario feasibility may involve: computational constraints, model capability, data availability, environment capability, time, physical interface limitations, or other constraints.

Where a materially required scenario cannot be simulated, the resulting Scenario Coverage Gap should remain visible.

Scenario Coverage

Scenario Coverage describes the extent to which the governed Scenario Set examines the materially relevant conditions required by the Simulation Purpose & Scope.

Coverage may include: normal conditions, variation, populations, environments, boundaries, uncertainty ranges, failure states, transitions, stresses, rare events, combinations, and recovery conditions.

Scenario Coverage ≠ Scenario Count

SIMULOS® establishes:

A large number of simulation scenarios does not establish adequate scenario coverage.

Ten thousand scenarios concentrated in one narrow region may provide weaker coverage than a smaller scenario set deliberately spanning all materially relevant condition classes.

Scenario Coverage Dimension

Coverage may be evaluated across multiple dimensions, such as:

Operating State

Environment

Population

Input Range

Uncertainty

Failure Mode

Transition

Stress

Interaction

Dependency

Recovery

or another simulation-specific dimension.

Coverage Matrix

Where useful, scenario coverage may be represented through a multidimensional matrix connecting:

Scenario Requirement × Condition Class × Scenario × Coverage Status

This allows gaps to become visible without reducing scenario completeness to one numerical count.

Scenario Coverage Status

A scenario requirement may receive a status such as:

Covered

Conditionally Covered

Partially Covered

Not Covered

or:

Coverage Undetermined Scenario Coverage Gap

A Scenario Coverage Gap exists where a materially relevant condition required by the Simulation Purpose & Scope is not sufficiently examined by the governed Scenario Set.

A gap may arise from: missing data, model limitation, environment limitation, computational constraints, unrecognized scenario need, excluded edge cases, missing combinations, or incomplete uncertainty coverage.

Coverage Gap Materiality

Scenario gaps should be assessed according to whether the omitted condition could materially affect: system behavior, outputs, risk understanding, failure understanding, operational decisions, or downstream validation and reliance.

Coverage Sufficiency

Scenario Coverage Sufficiency asks:

Does the Scenario Set provide enough governed variation and condition coverage to support the Simulation Purpose without claiming broader examination than was actually performed?

No Full-Coverage Claim Without Basis

SIMULOS® establishes:

Scenario coverage must not be represented as comprehensive where materially relevant scenario classes remain unexamined.

Scenario Redundancy

Multiple scenarios may examine substantially the same condition.

Redundancy may be useful for: repeatability, uncertainty exploration, random variation, or sensitivity testing.

But scenario count should not be inflated through redundant scenarios while important gaps remain.

Scenario Independence

Two scenarios should not be assumed independent where they depend upon the same: Input Set, Assumption, environment, failure source, behavioral model, or dependency.

This may matter when scenario results appear to provide independent confirmation.

Scenario Equivalence

Two scenarios may be scenario-equivalent where material differences do not alter the simulation inquiry.

Equivalence should be established rather than presumed.

Scenario Family

Related scenarios may be grouped into a Scenario Family.

Examples may include:

Communication Degradation Family

Power Failure Family

Extreme Environment Family

Autonomous Interaction Family

Scenario families may support systematic coverage without eliminating scenario-level identity.

Scenario Set

A Governed Simulation Scenario Set is the complete collection of scenarios approved for a defined simulation campaign or inquiry.

It may identify: scenario identities, scenario families, requirements,

configurations, applicable inputs, variation dimensions, edge conditions, stress conditions, coverage relationships, exclusions, gaps, priorities, and execution status.

Scenario Set Version

Scenario design may evolve.

A new Scenario Set Version should be distinguishable where materially relevant scenarios are: added, removed, modified, combined, reprioritized, or reclassified.

Scenario Change

Scenario change may be triggered by: new evidence, new input uncertainty, model change, environment change, discovered failure mode, changed Simulation Scope, new operational condition, changed assumptions, incident, or new system capability.

Scenario Reassessment

Material upstream change should trigger reassessment of whether existing scenario coverage remains sufficient.

No Static Scenario Set Principle

SIMULOS® establishes:

A Scenario Set that was sufficient for one simulation configuration or lifecycle state must not automatically be treated as sufficient after material change.

Scenario-to-Input Traceability

Scenarios should remain traceable to the input conditions they consume where material.

Conceptually:

Scenario → Simulation Input Set

Scenario-to-Parameter Traceability

Where scenarios differ through parameter configuration:

Scenario → Parameter Configuration

should remain traceable.

Scenario-to-Assumption Traceability

A scenario may exist specifically to test an assumption boundary or uncertainty.

Conceptually:

Simulation Assumption → Relevant Scenario Requirement → Scenario
Scenario-to-Environment Traceability

Environment-specific scenarios should remain connected to the Simulation
Environment Configuration they require.

Scenario-to-Execution Binding

Every Simulation Execution should be bindable to the scenario under which it occurred.

Conceptually:

Simulation Execution ID → Scenario ID

This allows outputs to remain traceable to the exact conditions examined.

Scenario Reproducibility

Where required, a scenario should be defined sufficiently to permit reconstruction of the materially relevant conditions under which it was executed.

Not every stochastic simulation will reproduce the identical output.

But the governed scenario configuration should remain reconstructable.

Scenario Comparability

Two scenario executions may be compared only where materially relevant differences are understood.

Scenario comparison should distinguish:

Intended Variation

from:

Unintended Configuration Difference.

Scenario Escalation

Scenario execution may reveal a condition that requires stronger or broader scenario coverage.

For example:

a nominal stress scenario may unexpectedly expose a cascading dependency.

Scenario governance should allow:

Observed Simulation Behavior → New Scenario Requirement → Scenario Set
Expansion

without altering completed historical records.

Scenario Discovery

Not every relevant scenario is known before simulation begins.

SIMULOS® therefore permits scenario discovery during the lifecycle.

Discovered scenarios should enter governance rather than remain informal observations.

Emergent Scenario

An Emergent Scenario is a materially relevant condition discovered through simulation interaction, execution behavior, external events, incident analysis, or new knowledge that was not part of the original Scenario Set.

Scenario Discovery Does Not Erase Earlier Gap

When an important scenario is discovered later, the architecture should preserve that earlier simulations did not examine it.

This distinction matters for historical reliance and reconstruction.

Scenario Selection Bias

Scenario design may become biased toward: expected successful behavior, easy-to-model conditions, familiar failure modes, available data, or scenarios supporting a desired conclusion.

SSS requires material selection logic to remain visible.

No Outcome-Seeking Scenario Principle

SIMULOS® establishes:

Scenario design must not be structured solely to produce the desired simulation outcome.

The Scenario Set should remain capable of exposing unfavorable, unexpected, or contradictory behavior relevant to the Simulation Purpose.

No Data-Convenience Scenario Principle

Scenario selection should not be limited exclusively to conditions for which convenient datasets already exist where material conditions outside those datasets remain within Simulation Scope.

No Model-Convenience Scenario Principle

A scenario does not cease to matter merely because the current Simulation Model cannot represent it sufficiently.

The result may instead be:

Scenario Required — Current Simulation Capability Insufficient Scenario
Capability Gap

A Scenario Capability Gap exists where the simulation architecture identifies a materially required scenario that cannot currently be represented or executed with sufficient governance.

The gap should remain explicit.

Scenario Readiness

A Scenario Set may receive a readiness status such as:

Scenario Ready

Conditionally Scenario Ready

Partially Scenario Ready

Scenario Not Ready

or:

Scenario Readiness Undetermined

Readiness applies to progression into governed Simulation Execution. Scenario
Readiness $\neq$ Simulation Validity

A Scenario Ready status means the scenario basis is sufficiently governed for execution.

It does not establish: correct execution, valid outputs, reality correlation,
or legitimate operational reliance.

Those questions belong to downstream SIMULOS® spaces and VALIDOS® where applicable.

Governance Boundary

SSS begins when the governed Simulation Object, Purpose & Scope, Reality Representation, Assumptions, Model & Environment, and Input & Data Basis are sufficiently established to determine which conditions the simulation should examine.

SSS ends when materially relevant scenarios have been sufficiently identified, varied, bounded, stressed, combined, prioritized, coverage-assessed, gap-documented, versioned, and assembled into a governed Simulation Scenario Set suitable for controlled Simulation Execution.

Architectural Boundary

SSS does not govern: general operational scenario planning, enterprise risk management, actual Operational Reality, general validation methodology, general integrity governance, Simulation Execution, Simulation Output governance, Reality Correlation, final simulation validation, or operational reliance.

SSS governs specifically:

the design and coverage of the conditions that SIMULOS® will ask the simulation to examine.

Minimum Governance Requirements

A conforming application of SSS should establish:

1. identifiable Scenario Requirements, 2. identifiable Simulation Scenarios, 3. baseline or reference scenarios where relevant, 4. expected-condition coverage, 5. material scenario variation, 6. edge-case identification, 7. boundary-condition coverage, 8. materially relevant stress conditions, 9. failure and degraded-state scenarios where applicable, 10. transition and recovery scenarios where applicable, 11. uncertainty-driven scenario treatment, 12. materially relevant scenario combinations, 13. scenario prioritization where exhaustive coverage is impractical, 14. explicit material scenario exclusions, 15. multidimensional Scenario Coverage assessment, 16. Scenario Coverage Gaps, 17. scenario-to-input/configuration traceability, 18. scenario versioning and change visibility, 19. Scenario Readiness, 20. and sufficient scenario-to-execution traceability for reconstruction.

Governance Outputs

SSS may produce: Scenario Requirement Register, Simulation Scenario Identifier, Simulation Scenario Record, Scenario Classification, Baseline Scenario, Nominal Scenario, Expected-Condition Scenario, Scenario Variation Record, Single-Factor Variation Record, Multi-Factor Variation Record, Edge-Case Scenario Record, Rare Scenario Record, Boundary Scenario Record, Boundary-Crossing Scenario Record, Stress Scenario Record, Extreme Scenario Record, Failure Scenario Record, Common-Cause Scenario Record, Cascading Scenario Record, Sequential Scenario Record, Transition Scenario Record, Degraded-State Scenario Record, Recovery Scenario Record, Uncertainty-Driven Scenario Record, Scenario Interaction Map, Scenario Dependency Map, Scenario Combination Record, Scenario Priority Record, Scenario Selection Basis, Scenario Exclusion Record, Scenario Capability Gap, Scenario Family, Scenario Coverage Matrix, Scenario Coverage Assessment, Scenario Coverage Status, Scenario Coverage Gap, Scenario Coverage Gap Materiality Assessment, Scenario Coverage Sufficiency Assessment, Scenario Set, Scenario Set Identifier, Scenario Set Version, Scenario Change Record, Scenario Reassessment Trigger, Scenario Discovery Record, Emergent Scenario Record, Scenario-to-Assumption Traceability Reference, Scenario-to-Input Traceability Map, Scenario-to-Parameter Traceability Map, Scenario-to-Environment Traceability Map, Scenario-to-Execution Binding Requirement, Scenario Readiness Status, and Governed Simulation Scenario Set.

Relationship to Other OOF® Architectures

SSS remains bounded to simulation-specific Scenario Design.

ORA™ governs actual Operational Reality. SSS may use operational conditions as scenario references but does not govern real-world operational states themselves.

VALIDOS® governs validation. Scenario coverage produced under SSS may later become evidence relevant to validation, but SSS does not itself determine whether the simulation or its results satisfy a formal Validation Determination.

INTEGROS® may govern integrity of scenario definitions, records, configurations, and preserved scenario history.

ADIT® may preserve evidence of scenario design, selection, change, execution, and coverage.

AGA™ may govern accountability for material scenario selection, exclusion, prioritization, or approval decisions.

ASGA™ may define autonomous-system governance conditions that become scenario requirements when autonomous systems are simulated.

AIG® may contribute AI behavioral conditions that require scenario representation.

SSS does not replace those architectures.

It governs specifically which simulation conditions must be deliberately constructed and examined before simulation execution can claim meaningful scenario coverage.

Architectural Position

Simulation Scenario Standard (SSS) is the seventh Parent Standard of SIMULOS®
— Simulation Governance Architecture.

Its position in the 12-Parent-Standard architecture is:

1. Simulation Object

↓

2. Simulation Purpose & Scope

↓

3. Reality Representation

↓

4. Simulation Assumptions

↓

5. Simulation Model & Environment



6. Simulation Inputs & Data



7. Simulation Scenario Design



8. Simulation Execution



9–12. Downstream SIMULOS® lifecycle governance

SSS occupies the point where the previously governed simulation foundation becomes an explicit testable condition space.

Before SSS, SIMULOS® establishes:

what is being simulated, why, how reality is represented, what is assumed, through which model and environment the simulation operates, and which information may enter it.

SSS then determines:

which materially relevant conditions that governed simulation must actually examine.

Only after that scenario basis exists should the architecture proceed into controlled Simulation Execution.

Foundational Principle

A simulation cannot claim meaningful coverage merely because it executes successfully; it must deliberately examine the materially relevant conditions, variations, boundaries, uncertainties, failures, stresses, interactions, and alternatives capable of changing the behavior or conclusions of the simulation.

Canonical Closing Statement

Simulation Scenario Standard establishes the scenario-design governance layer of SIMULOS® by defining how materially relevant simulation conditions are identified, structured, varied, combined, stressed, bounded, prioritized, coverage-assessed, and preserved before execution. By governing nominal conditions, expected variation, edge cases, boundary conditions, rare events, degraded and failure states, transitions, cascading interactions, uncertainty-driven scenarios, stress conditions, scenario exclusions, coverage gaps, scenario readiness, versioning, and complete scenario-to-execution traceability, SSS prevents simulation volume from being mistaken for simulation coverage and ensures that a simulation cannot appear comprehensively examined while the conditions most capable of changing its

behavior remain outside the governed Scenario Set.

Module Architecture

→ [Scenario Identification Module \(SCIM\)](#).

→ [Scenario Variation Module \(SCVM\)](#).

→ [Scenario Edge-Case Module \(SECM\)](#).

→ [Scenario Stress-Condition Module \(SSCM\)](#).

→ [Scenario Coverage Module \(SCCM\)](#).

Related Documents

→ [About Simulation Scenario Standard \(SSS\)](#).

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>